COMPOSITION OF TWO FLAT BASE CHANGES IS A FLAT BASE CHANGE AND THE PULLBACKS COMPOSE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0FVP (sources/stacks-chow/pinned-original-source.tex line 15438).

Allowed standard prerequisites (editorial).

- The base change situation and its shift constant. Same chapter, Situation labelled situation-setup-base-change, invoked in the statement. Fixes the hypotheses and the meaning of the shifts c and c'.
- Lifting of generalizations along flat morphisms. Morphisms, Lemma labelled morphisms-lemma-base-change-flat, cited at proof lines 15470-15472. Makes the image of a generic point of a fibre component again a generic point of a fibre component.
- Base change of cycles and its composition. Same chapter, Lemma labelled lemma-pullback-coherent-base-change, cited at proof lines 15478-15480. Gives (g')^*[Z']_{k+c} = [Z"]_{k+c+c'} and hence the composition on Chow groups.
- The pullback map on Chow groups. Same chapter, Lemma labelled lemma-pullback-base-change, cited in the statement. Identifies the Chow level maps whose composition is asserted.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Same-chapter pointer into the author environment situation at source line 1372; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original reference 0FVG: sources/stacks-chow/pinned-original-source.tex, source lines 15123–15123. Same-chapter pointer into the author environment situation at source line 15123; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Date: stacks-project:chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0FVJ: sources/stacks-chow/pinned-original-source.tex, source lines 15255–15255. Same-chapter pointer into the author environment lemma at source line 15255; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 01U9: sources/stacks-chow/tags.txt, source lines 2402–2402. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 15467 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FVI: sources/stacks-chow/pinned-original-source.tex, source lines 15193–15193. Same-chapter pointer into the author environment lemma at source line 15193; invoked at proof line 15478 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let (S, δ), (S', δ'), (S'', δ'') be as in Situation 02QL. Let $g : S' \to S$ and $g' : S'' \to S'$ be flat morphisms of schemes and let $c, c' \in \mathbf{Z}$ be integers such that $S, \delta, S', \delta', g, c$ and S', δ', S'', g', c' are as in Situation 0FVG. Let $X \to S$ be locally of finite type and denote $X' \to S'$ and $X'' \to S''$ the base changes by $S' \to S$ and $S'' \to S$. Then*

1. *$S, \delta, S'', \delta'', g \circ g', c + c'$ is as in Situation 0FVG,*
2. *the maps $g^* : Z_k(X) \to Z_{k+c}(X')$ and $(g')^* : Z_{k+c}(X') \to Z_{k+c+c'}(X'')$ of compose to give the map $(g \circ g')^* : Z_k(X) \to Z_{k+c+c'}(X'')$, and*
3. *the maps $g^* : \mathrm{CH}_k(X) \to \mathrm{CH}_{k+c}(X')$ and $(g')^* : \mathrm{CH}_{k+c}(X') \to \mathrm{CH}_{k+c+c'}(X'')$ of Lemma 0FVJ compose to give the map $(g \circ g')^* : \mathrm{CH}_k(X) \to \mathrm{CH}_{k+c+c'}(X'')$ of Lemma 0FVJ.*

Proof. Let $s \in S$ and let $s'' \in S''$ be a generic point of an irreducible component of $(g \circ g')^{-1}(\{s\})$. Set $s' = g'(s'')$. Clearly, s'' is a generic point of an irreducible component of $(g')^{-1}(\{s'\})$. Moreover, since g' is flat and hence generalizations lift along g' (Morphisms, Lemma 01U9) we see that also s' is a generic point of an irreducible component of $g^{-1}(\{s\})$. Thus by assumption $\delta'(s') = \delta(s) + c$ and $\delta''(s'') = \delta'(s') + c'$. We conclude $\delta''(s'') = \delta(s) + c + c'$ and the first part of the statement is true.

For the second part, let $Z \subset X$ be an integral closed subscheme of δ-dimension k. Denote $Z' \subset X'$ and $Z'' \subset X''$ the base changes. By definition we have $g^*[Z] = [Z']_{k+c}$. By Lemma 0FVI we have $(g')^*[Z']_{k+c} = [Z'']_{k+c+c'}$. This proves the final statement. $\square$